

Titration report



Method

Method ID	SERCALKSAMPLES	Release state	Released
Name	TA of Seawater	Modified by	SystemInternal
Modified on	2/22/2024 12:07:06 PM		

Summary

Samples

	Sample ID	Name	Result	State
1	FTS2023 8/81630 rep3, 16	Sample size	59.819 g	
		Content R1	1.9442 mmol/L	OK
2	FTS2023 8/8 1630 rep4, 17	Sample size	62.129 g	
		Content R1	1.9466 mmol/L	OK

Statistics

R1

Minimum	1.9454 mmol/L	Minimum	1.9442 mmol/L
Standard deviation	0.0017	Rel. standard deviation	0.087 %
Total samples	2	Samples excluded	0

Main Result Details

Result title	Content	Sample ID1	FTS2023 8/81630 rep3
Result name	R1	Sample ID2	16
Result	1.9442 mmol/L	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Result title	Content	Sample ID1	FTS2023 8/8 1630 rep4
Result name	R1	Sample ID2	17
Result	1.9466 mmol/L	Result comment	
		Result state	OK
		Result state comment	
		Nominal value	
		Compliance state	Released
		Is result deleted	No

Calculation formula R1=Q*C/m

Titration report

LabX

Raw Results

Scope 1/1, Sample 1/2 (FTS2023 8/81630 rep3; 16)

Sample (S1)

Creation time	2/22/2024 6:25:26 PM
Sample start time	2/22/2024 6:25:26 PM
Sample size	59.819 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 6:25:27 PM
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Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 6:25:27 PM
T	25.0 °C
E	7.775 pH
tMe	01:16 min
ESTMe	4.598 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 6:26:43 PM
Terminated at	Potential
EEQ1	4.572 pH ⁽⁹⁾
EHN1	6.234 pH
Q1	0.116303 mmol ⁽⁹⁾
VEQ1	1.161870 mL ⁽⁹⁾
EST	7.781 pH
VEX	0.824630 mL
VEND	1.98650 mL
t	23:22 min
QEND	0.198849 mmol
QEX	0.082545 mmol
EPD	6.662 pH
nEQ	1

Content (R1)

Creation time	2/22/2024 6:50:12 PM
Result	1.9442 mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 6:50:13 PM
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Scope 1/1, Sample 2/2 (FTS2023 8/8 1630 rep4; 17)

Titration report

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Sample (S1)

Creation time	2/22/2024 6:50:23 PM
Sample start time	2/22/2024 6:50:23 PM
Sample size	62.129 g
Correction factor	1.0
Temperature	25.0 °C
Density	1.0 g/mL

Titration stand (Titrationstand1)

Creation time	2/22/2024 6:50:24 PM
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Measure (normal) [1] (Measurenormal1)

Creation time	2/22/2024 6:50:24 PM
T	25.1 °C
E	7.813 pH
tMe	01:12 min
ESTMe	4.765 pH

Titration (EQP) [1] (TitrationEQP1)

Creation time	2/22/2024 6:51:36 PM
Terminated at	Potential
EEQ1	4.563 pH ⁽⁹⁾
EHN1	6.235 pH
Q1	0.120939 mmol ⁽⁹⁾
VEQ1	1.208181 mL ⁽⁹⁾
EST	7.818 pH
VEX	0.842819 mL
VEND	2.05100 mL
t	23:51 min
QEND	0.205305 mmol
QEX	0.084366 mmol
EPD	6.687 pH
nEQ	1

Content (R1)

Creation time	2/22/2024 7:15:33 PM
Result	1.9466 mmol/L
Decimal places	4
Substance	None
M	1 g/mol
z	1

Rinse (Rinse1)

Creation time	2/22/2024 7:15:33 PM
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After Sample Scopes

Park (Park1)

Creation time	2/22/2024 7:15:46 PM
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Measure (normal) [1] Scope 1/1, Sample 1/2

Measured Values Scope 1/1, Sample 1/2

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.598	19.3	
2	5.020	20.0	0.422
3	5.289	20.8	0.269
4	5.483	21.7	0.194
5	5.642	22.4	0.159
6	5.783	23.0	0.141
7	5.915	23.4	0.132
8	6.027	23.8	0.112
9	6.139	24.0	0.112
10	6.244	24.3	0.105
11	6.345	24.4	0.101
12	6.439	24.6	0.094
13	6.528	24.7	0.089
14	6.612	24.7	0.084
15	6.682	24.8	0.070
16	6.755	24.8	0.073
17	6.826	24.9	0.071
18	6.892	24.9	0.066
19	6.953	24.9	0.061
20	7.008	25.0	0.055
21	7.055	25.0	0.047
22	7.110	25.0	0.055
23	7.158	25.0	0.048
24	7.201	25.0	0.043
25	7.242	25.0	0.041
26	7.282	25.0	0.040
27	7.316	25.0	0.034
28	7.351	25.0	0.035
29	7.383	25.0	0.032
30	7.414	25.0	0.031
31	7.442	25.0	0.028
32	7.467	25.0	0.025
33	7.492	25.0	0.025
34	7.512	25.0	0.020
35	7.533	25.0	0.021
36	7.554	25.0	0.021
37	7.572	25.0	0.018
38	7.590	25.0	0.018
39	7.606	25.0	0.016
40	7.620	25.0	0.014
41	7.633	25.0	0.013
42	7.647	25.0	0.014
43	7.659	25.0	0.012
44	7.670	25.0	0.011
45	7.680	25.0	0.010

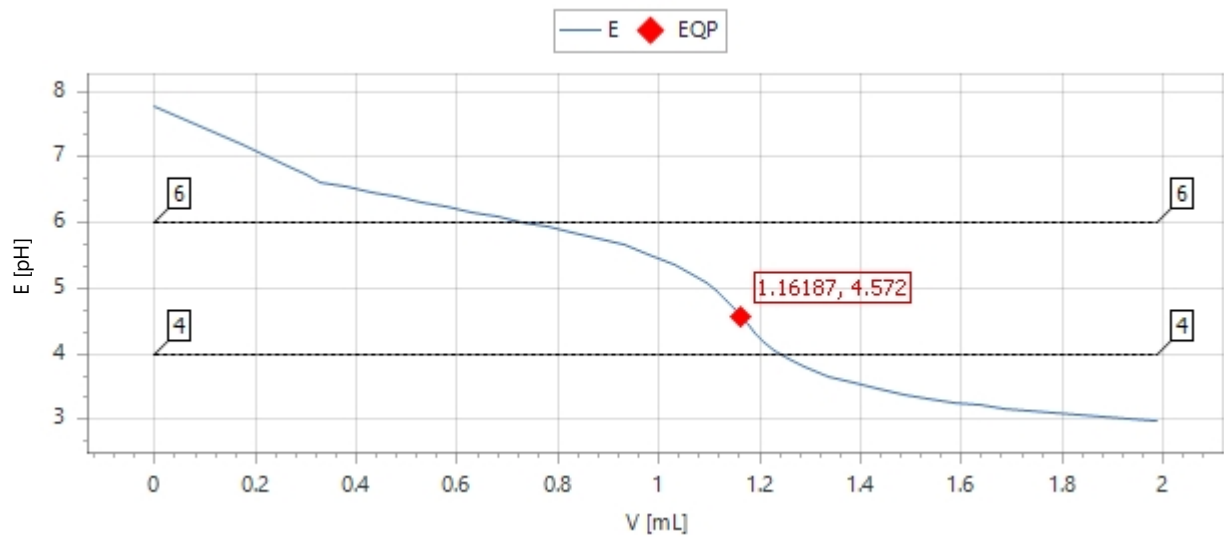
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LabX

46	7.690	25.0	0.010
47	7.698	25.0	0.008
48	7.707	25.0	0.009
49	7.714	25.0	0.007
50	7.722	25.0	0.008
51	7.729	25.0	0.007
52	7.736	25.0	0.007
53	7.742	25.0	0.006
54	7.747	25.0	0.005
55	7.752	25.0	0.005
56	7.757	25.0	0.005
57	7.762	25.0	0.005
58	7.767	25.0	0.005
59	7.771	25.0	0.004
60	7.774	25.0	0.003

Titration (EQP) [1] Scope 1/1, Sample 1/2

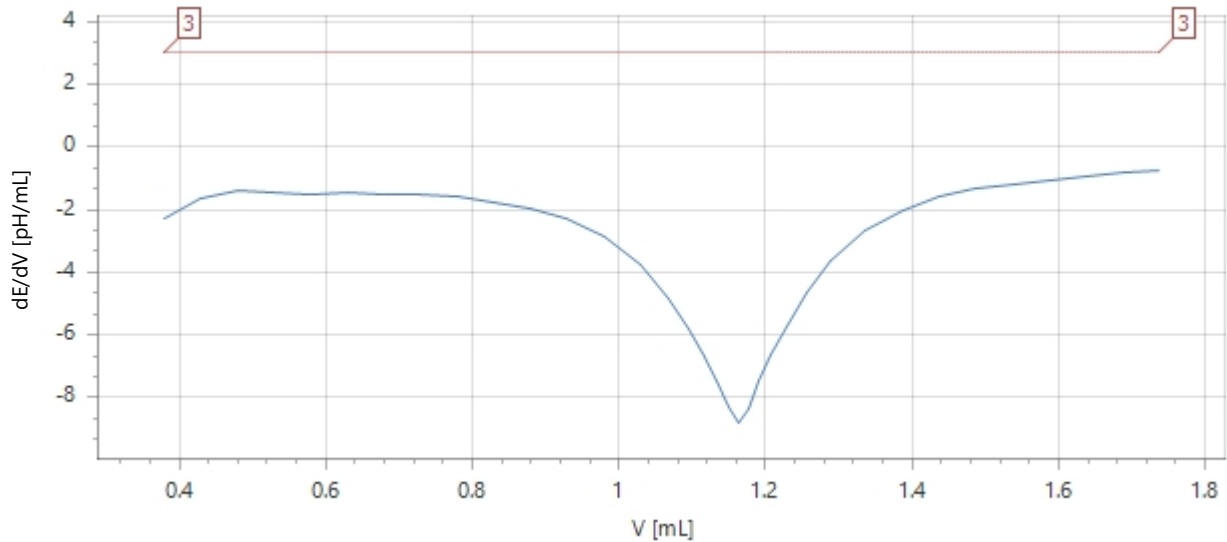
E - V Curve Scope 1/1, Sample 1/2



dE/dV - V Curve Scope 1/1, Sample 1/2

Titration report

LabX



Measured Values Scope 1/1, Sample 1/2

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.781		25.0			
30	0.17150	7.203	0.17150	25.0	-0.578		
60	0.25725	6.894	0.08575	25.0	-0.309		
90	0.30000	6.722	0.04275	25.0	-0.172		
181	0.32975	6.623	0.02975	25.0	-0.099		
211	0.37975	6.547	0.05000	25.0	-0.076	-2.27	
241	0.42975	6.466	0.05000	25.0	-0.081	-1.63	
271	0.47975	6.389	0.05000	25.0	-0.077	-1.40	
302	0.52975	6.312	0.05000	25.0	-0.077	-1.46	
332	0.57975	6.236	0.05000	25.0	-0.076	-1.49	
362	0.62975	6.165	0.05000	25.0	-0.071	-1.48	
392	0.67975	6.091	0.05000	25.0	-0.074	-1.49	
422	0.72975	6.011	0.05000	25.0	-0.080	-1.53	
453	0.77975	5.936	0.05000	25.0	-0.075	-1.61	
483	0.82975	5.850	0.05000	25.0	-0.086	-1.74	
513	0.87975	5.758	0.05000	25.0	-0.092	-1.95	
544	0.92975	5.652	0.05000	25.0	-0.106	-2.30	
574	0.97975	5.527	0.05000	25.0	-0.125	-2.87	
604	1.02975	5.372	0.05000	25.0	-0.155	-3.78	
634	1.06775	5.220	0.03800	25.0	-0.152	-4.84	
664	1.09400	5.089	0.02625	25.0	-0.131	-5.79	
695	1.11575	4.954	0.02175	25.0	-0.135	-6.68	
725	1.13350	4.823	0.01775	25.0	-0.131	-7.47	
755	1.14900	4.691	0.01550	25.0	-0.132	-8.30	
	1.161870 ⁽⁹⁾	4.572 ⁽⁹⁾					EQP
785	1.16275	4.564	0.01375	25.0	-0.127	-8.78	
816	1.17625	4.440	0.01350	25.0	-0.124	-8.37	
846	1.19125	4.311	0.01500	25.0	-0.129	-7.47	
876	1.20800	4.185	0.01675	25.0	-0.126	-6.65	
906	1.22850	4.060	0.02050	25.0	-0.125	-5.75	
937	1.25600	3.925	0.02750	25.0	-0.135	-4.66	

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967	1.29025	3.795	0.03425	25.0	-0.130	-3.62
997	1.33650	3.660	0.04625	25.0	-0.135	-2.66
1027	1.38650	3.550	0.05000	25.0	-0.110	-2.01
1058	1.43650	3.461	0.05000	25.0	-0.089	-1.61
1088	1.48650	3.386	0.05000	25.0	-0.075	-1.35
1118	1.53650	3.323	0.05000	25.0	-0.063	-1.17
1148	1.58650	3.267	0.05000	25.0	-0.056	-1.04
1178	1.63650	3.217	0.05000	25.0	-0.050	-0.93
1209	1.68650	3.173	0.05000	25.0	-0.044	-0.84
1239	1.73650	3.133	0.05000	25.0	-0.040	-0.77
1269	1.78650	3.095	0.05000	25.0	-0.038	
1300	1.83650	3.062	0.05000	25.0	-0.033	
1330	1.88650	3.030	0.05000	25.0	-0.032	
1360	1.93650	3.001	0.05000	25.0	-0.029	
1390	1.98650	2.973	0.05000	25.0	-0.028	

Measure (normal) [1] Scope 1/1, Sample 2/2

Measured Values Scope 1/1, Sample 2/2

Time [s]	Meas. value [pH]	Temp. [°C]	dE [pH]
1	4.765	23.1	
2	5.040	23.3	0.275
3	5.288	23.6	0.248
4	5.489	23.9	0.201
5	5.658	24.2	0.169
6	5.805	24.4	0.147
7	5.938	24.5	0.133
8	6.052	24.6	0.114
9	6.172	24.7	0.120
10	6.282	24.8	0.110
11	6.386	24.9	0.104
12	6.483	24.9	0.097
13	6.574	25.0	0.091
14	6.662	25.0	0.088
15	6.737	25.0	0.075
16	6.814	25.0	0.077
17	6.888	25.1	0.074
18	6.953	25.1	0.065
19	7.012	25.1	0.059
20	7.071	25.1	0.059
21	7.128	25.1	0.057
22	7.174	25.1	0.046
23	7.220	25.1	0.046
24	7.265	25.1	0.045
25	7.307	25.1	0.042
26	7.346	25.1	0.039
27	7.381	25.1	0.035
28	7.415	25.1	0.034
29	7.442	25.1	0.027

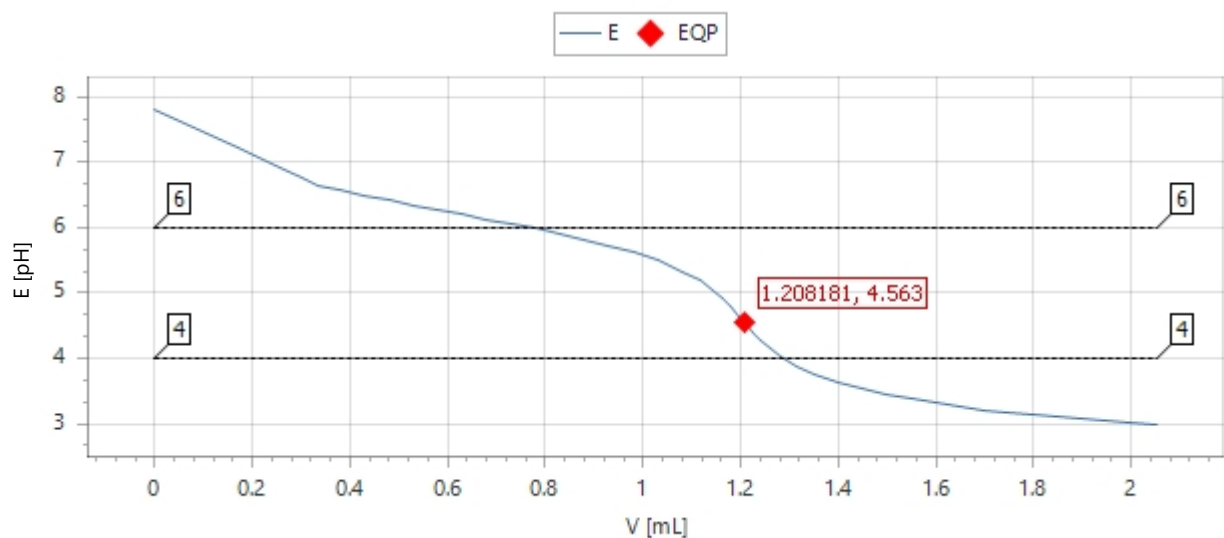
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30	7.471	25.1	0.029
31	7.498	25.1	0.027
32	7.522	25.1	0.024
33	7.545	25.1	0.023
34	7.568	25.1	0.023
35	7.588	25.1	0.020
36	7.606	25.1	0.018
37	7.624	25.1	0.018
38	7.640	25.1	0.016
39	7.655	25.1	0.015
40	7.669	25.1	0.014
41	7.682	25.1	0.013
42	7.694	25.1	0.012
43	7.705	25.1	0.011
44	7.716	25.1	0.011
45	7.726	25.1	0.010
46	7.735	25.1	0.009
47	7.744	25.1	0.009
48	7.752	25.1	0.008
49	7.758	25.1	0.006
50	7.765	25.1	0.007
51	7.771	25.1	0.006
52	7.777	25.1	0.006
53	7.783	25.1	0.006
54	7.788	25.1	0.005
55	7.793	25.1	0.005
56	7.797	25.1	0.004
57	7.801	25.1	0.004
58	7.805	25.1	0.004
59	7.809	25.1	0.004
60	7.812	25.1	0.003

Titration (EQP) [1] Scope 1/1, Sample 2/2

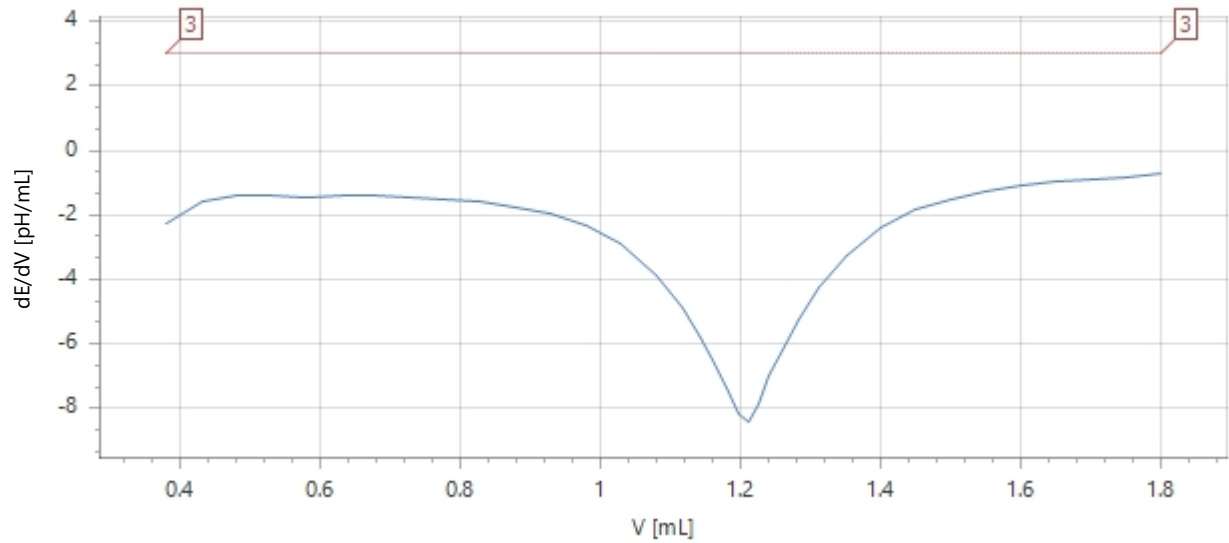
E - V Curve Scope 1/1, Sample 2/2



Titration report

LabX

dE/dV - V Curve Scope 1/1, Sample 2/2



Measured Values Scope 1/1, Sample 2/2

Time [s]	V [mL]	Meas. value [pH]	dV [mL]	Temp. [°C]	dE [pH]	dE/dV [pH/mL]	EQP
0	0.00000	7.818		25.1			
28	0.17150	7.242	0.17150	25.1	-0.576		
58	0.25725	6.922	0.08575	25.1	-0.320		
89	0.30000	6.753	0.04275	25.1	-0.169		
180	0.33300	6.643	0.03300	25.0	-0.110		
210	0.38050	6.571	0.04750	25.0	-0.072	-2.28	
240	0.43050	6.495	0.05000	25.0	-0.076	-1.61	
270	0.48050	6.417	0.05000	25.0	-0.078	-1.38	
300	0.53050	6.341	0.05000	25.0	-0.076	-1.41	
331	0.58050	6.267	0.05000	25.0	-0.074	-1.44	
361	0.63050	6.200	0.05000	25.0	-0.067	-1.41	
391	0.68050	6.127	0.05000	25.0	-0.073	-1.41	
422	0.73050	6.057	0.05000	25.0	-0.070	-1.45	
452	0.78050	5.983	0.05000	25.0	-0.074	-1.51	
482	0.83050	5.903	0.05000	25.1	-0.080	-1.60	
512	0.88050	5.818	0.05000	25.1	-0.085	-1.74	
542	0.93050	5.726	0.05000	25.1	-0.092	-1.97	
573	0.98050	5.620	0.05000	25.1	-0.106	-2.32	
603	1.03050	5.494	0.05000	25.1	-0.126	-2.90	
633	1.08050	5.335	0.05000	25.1	-0.159	-3.86	
664	1.11650	5.190	0.03600	25.1	-0.145	-4.88	
694	1.14350	5.052	0.02700	25.1	-0.138	-5.83	
724	1.16400	4.922	0.02050	25.1	-0.130	-6.62	
754	1.18125	4.795	0.01725	25.1	-0.127	-7.36	
784	1.19725	4.660	0.01600	25.1	-0.135	-8.16	
	1.208181 ⁽⁹⁾	4.563 ⁽⁹⁾					EQP
815	1.21150	4.533	0.01425	25.1	-0.127	-8.40	
845	1.22575	4.410	0.01425	25.1	-0.123	-7.84	
875	1.24175	4.282	0.01600	25.1	-0.128	-6.98	

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905	1.26025	4.154	0.01850	25.1	-0.128	-6.18
936	1.28300	4.026	0.02275	25.1	-0.128	-5.26
966	1.31275	3.893	0.02975	25.1	-0.133	-4.23
996	1.35100	3.763	0.03825	25.1	-0.130	-3.24
1026	1.40100	3.632	0.05000	25.1	-0.131	-2.37
1057	1.45100	3.530	0.05000	25.1	-0.102	-1.84
1087	1.50100	3.447	0.05000	25.1	-0.083	-1.50
1117	1.55100	3.378	0.05000	25.1	-0.069	-1.28
1147	1.60100	3.317	0.05000	25.1	-0.061	-1.11
1178	1.65100	3.264	0.05000	25.1	-0.053	-0.99
1208	1.70100	3.216	0.05000	25.0	-0.048	-0.89
1238	1.75100	3.174	0.05000	25.1	-0.042	-0.81
1268	1.80100	3.135	0.05000	25.1	-0.039	-0.74
1299	1.85100	3.099	0.05000	25.1	-0.036	
1329	1.90100	3.066	0.05000	25.1	-0.033	
1359	1.95100	3.035	0.05000	25.1	-0.031	
1389	2.00100	3.006	0.05000	25.0	-0.029	
1419	2.05100	2.980	0.05000	25.0	-0.026	

Used Instruments

Name	SERCtitrate	Serial number	B433917554
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Responsible User

Time	Login name	Full name
2/22/2024 6:25:25 PM	SystemInternal	System internal account

(9) Standard evaluation used